



# Assignment 1 Report

Variables, Functions, LEDs, while Loops & Serial Monitor

**Name:** Benjamin Agalisi  
**Course:** Programming C++ for Engineers Using Arduino  
**Program of Study:** BSc Computer Engineering

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This report summarizes the five tasks completed in Assignment 1, covering Arduino fundamentals: variables, functions, digital LED output, while loops, and Serial Monitor communication.

## Task 1 — Hello, Serial!

Created variables of different data types (int, float, String, bool) and printed them to the Serial Monitor using `Serial.print()` and `Serial.println()`. This introduced the difference between `print` (no newline) and `println` (adds a newline), and confirmed `Serial.begin(9600)` is required before any Serial communication will work.

## Task 2 — First Blink Using Variables

Controlled the built-in LED (pin 13) using variables (`ledPin`, `onTime`, `offTime`) instead of hardcoded values. Demonstrated that `pinMode()` belongs in `setup()` (runs once) while the blink logic belongs in `loop()` (runs repeatedly). Changing the timing variables alone was enough to change the blink speed, without touching the loop logic.

## Task 3 — Make It a Function

Refactored the blink logic into a custom function, `blinkOnce(int waitTime)`, called three times from `loop()` with different wait values (fast, medium, slow). This showed how functions let repeating logic be written once and reused, keeping `loop()` clean and readable.

## Task 4 — Countdown with a while Loop

Used a while loop to count down from 5 to 1, printing each value to the Serial Monitor with a 1-second delay between numbers, followed by three celebratory LED flashes. This task highlighted the importance of updating the loop variable (`counter = counter - 1`) to avoid an infinite loop.

## Task 5 — Smart Countdown (Combined Program)

Combined all previous concepts into one program: a `startValue` variable, a custom `flashLED(int times)` function, and a while loop that counts down from 5 to 1. On each step the program prints the count, blinks the LED that many times, then waits 1 second, finishing with a completion message on the Serial Monitor.

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## Key Concepts Demonstrated

Variables and data types (int, float, bool, String) · Defining and calling custom functions with parameters · Digital output with pinMode() and digitalWrite() · Controlled repetition with while loops · Serial communication for debugging and output (Serial.begin, Serial.print, Serial.println).

## Reflection

Working through these tasks built a solid foundation in core Arduino programming concepts. The progression from simple variable printing to a fully combined program (Task 5) showed how small, well-tested building blocks can be assembled into a complete, working system — a pattern that applies broadly to engineering and software development.

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Submitted by Benjamin Agalisi — BSc Computer Engineering

